



University of Asia Pacific

Department of Computer Science and Engineering

CSE 450: VLSI Lab

LAB REPORT

Experiment Number: 04

Experiment Title: Design and Simulation of an 8-Bit Register Using D Flip-Flops

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1. Title of the Experiment

Design and Simulation of an 8-Bit Register Using D Flip-Flops

2. Objectives

The following are the experiment's goals:

1. To use structural Verilog to design an 8-bit register.
2. To use flip-flop modules to build the register.
3. To comprehend Verilog structural and hierarchical design.
4. To use Xilinx ISE to generate and examine the 8-bit register's RTL schematic.
5. To use a testbench to mimic the designed register and confirm its operation.
6. To comprehend how a multi-bit storage element can be built by combining individual flip-flops.

3. Software and Tools Used

The following tools were used for this experiment:

- Xilinx ISE 14.7
- Verilog
- ISim Simulator
- Computer system for design and simulation

4. Design Methodology

The 8-bit register was designed using a hierarchical structural approach.

First, a basic 1-bit register was designed using a D flip-flop. The 1-bit register stores one input bit and updates its output on the rising edge of the clock.

Then, the 8-bit register was constructed by instantiating eight 1-bit register modules. Each module stores one bit of the 8-bit data, and all eight registers share the same clock signal.

The basic hierarchy is:

D Flip-Flop → 1-Bit Register → 8-Bit Register

The complete 8-bit register consists of:

- **REG0** – stores Bit 0
- **REG1** – stores Bit 1
- **REG2** – stores Bit 2
- **REG3** – stores Bit 3
- **REG4** – stores Bit 4
- **REG5** – stores Bit 5
- **REG6** – stores Bit 6
- **REG7** – stores Bit 7

Finally, a **testbench** was created to apply different 8-bit input values and verify the register operation through **ISim behavioral simulation**.

5. RTL-Schematic Diagrams

The top-level RTL schematic represents the complete **8-bit register** and shows the structural connection of the eight 1-bit register modules.

The 8-bit register is constructed using:

- **REG0–REG3** – store the lower 4 bits
- **REG4–REG7** – store the upper 4 bits

Each 1-bit register receives its corresponding data input and shares the same **clock signal**. The outputs of the eight 1-bit registers form the complete 8-bit registered output.

The same structural design is used for all eight register instances, with each instance responsible for storing one individual bit.

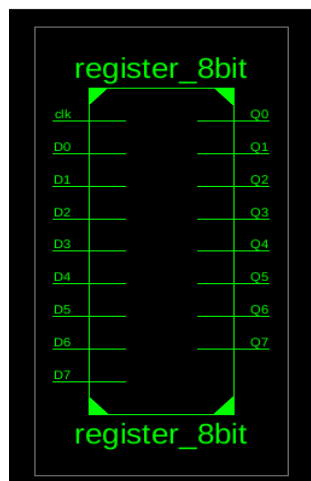


Fig: RTL schematic diagram of 8 bit register

6. GitHub Repository

All Verilog source codes, testbenches, and related project files should be uploaded to a GitHub repository.

GitHub Repository Link

<https://github.com/Akifhossain29/ISE-14.7-VLSI>

7. Simulation

The designed 8-bit register was simulated using Xilinx ISE ISim with a Verilog testbench.

The testbench applies different 8-bit input values to the register and observes the corresponding outputs after the clock transitions.

The main purpose of the simulation is to verify that:

- The register correctly accepts 8-bit input data.
- Each 1-bit register stores its corresponding input bit.
- The stored data appears correctly at the outputs.
- All eight register modules operate with the common clock signal.
- The complete hierarchical structural design works as expected.

7.1 Simulation Waveform

The simulation waveform demonstrates different 8-bit input values being applied to the register and shows how the corresponding output values are stored after the active clock edge.

For example:

Input: 10101010

↓

Register: Stores the input at the rising edge of the clock

↓

Output: 10101010

Another input value is then applied to verify that the register updates correctly on the next rising clock edge.

The simulation was performed using Xilinx ISE → ISim. The waveform shows:

- Clock signal
- 8-bit input signals (D0–D7)
- 8-bit output signals (Q0–Q7)

The waveform is used to verify that each 1-bit register correctly stores its corresponding input bit and that all eight registers operate with the common clock signal.

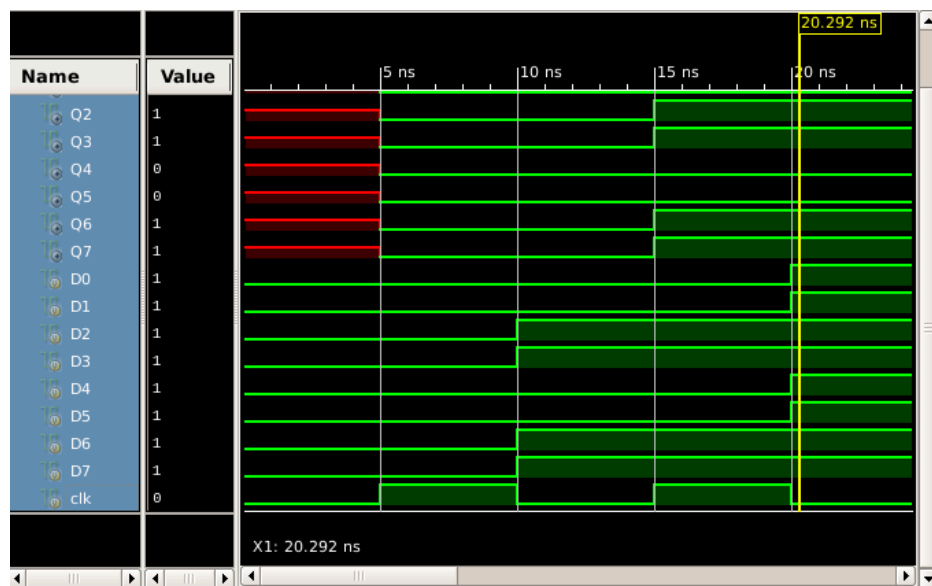


Fig: 8 bit register waveform

8. Simulation Analysis

The simulation waveform was analyzed by comparing the input and output signals at the rising edges of the clock.

When a valid 8-bit input is applied, the eight 1-bit registers store their corresponding input bits at the rising edge of the clock. The stored data is then observed at the output signals.

For example, in the simulation, when the input is:

$D[7:0] = 10101010$

the register stores the data and produces the corresponding output:

$Q[7:0] = 10101010$

Similarly, other input patterns were applied during the simulation, and the output changed according to the input data at the appropriate clock edge.

Therefore, the simulation verifies that the complete 8-bit register correctly stores and updates binary data according to the clock signal.

12. Conclusion

In this experiment, an 8-bit register was designed and simulated using structural Verilog in Xilinx ISE. The register was constructed using a hierarchical structural approach.

First, a 1-bit register was designed using a D flip-flop. Then, eight 1-bit register modules were instantiated to construct the complete 8-bit register. Each register stores one input bit, and all eight registers share the same clock signal.

The RTL schematic and module hierarchy demonstrated the structural organization of the design, while the ISim simulation waveform verified its functional behavior with different 8-bit input values.

The experiment provided practical understanding of hierarchical structural Verilog design, D flip-flop-based registers, 8-bit register organization, RTL structure, and digital circuit simulation.